Introduction to Discrete Probability

Section 7.1

Video 73: Probability

- Finite probability
- Examples

Probability of an Event

Pierre-Simon Laplace
(1749-1827)

Pierre-Simon Laplace's classical theory of probability was introduced in the 18th century, when he analyzed games of chance.

Definitions:

- An **experiment** is a procedure that yields one of a given set of possible **outcomes**: rolling the dice
- The **sample space S** of the experiment is the set of possible outcomes: faces of the dice S = {1,2,3,4,5,6}
- An **event E** is a subset of the sample space: e.g., E = {5,6}
- An event **occurs**, if the outcome belongs to the sample space: E occurs when rolling a 6

Probability of an Event

Definition: If S is a finite sample space of equally likely outcomes, and E is an event, that is, a subset of S, then the **probability** of E is

$$p(E) = \frac{|E|}{|S|}$$

For every event E, we have $0 \leq p(E) \leq 1$.

This follows directly from the definition because

$$0 \leq p(E) = |E|/|S| \leq |S|/|S| \leq 1, \text{ since } 0 \leq |E| \leq |S|.$$

Example

What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

- By the product rule there are $6^2 = 36$ possible outcomes
- 6 of these sum up to 7: 1+6, 2+5, 3+4, 4+3, 5+2, 6+1
- Hence, the probability of obtaining a 7 is 6/36 = 1/6

Example

There are many lotteries that award prizes to people who correctly choose <u>a set of six numbers</u> out of the first n positive integers, where n is usually between 30 and 60.

What is the probability that a person picks the correct six numbers out of 40?

- The number of ways to choose six numbers out of 40 is

$$C(40,6) = 40!/(34!6!) = 3{,}838{,}380.$$

- Hence, the probability of picking a winning combination is $1/3{,}838{,}380 \approx 0.00000026$.

Example

What is the probability that the numbers 11, 4, 17, 39, and 23 are drawn <u>in that order</u> from a bin with 50 balls labeled with the numbers 1,2, …, 50 if

1. The ball selected is not returned to the bin.
2. The ball selected is returned to the bin before the next ball is selected.

Use the product rule in each case.

1. *Sampling without replacement*: The probability is 1/254,251,200 since there are $50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 = 254{,}251{,}200$ ways to choose the five balls.
2. *Sampling with replacement*: The probability is $1/50^5 = 1/312{,}500{,}000$ since $50^5 = 312{,}500{,}000$.

Summary

- Finite probability
  - Sample Space, Event, Probability
- Examples
  - Card Games, Lotteries

Video 74: Complements and Unions of Events

- Complements
- Unions
- Examples

The Probability of Complements of Events

Theorem 1: Let *E* be an event in sample space *S*. The probability of the event $\bar{E} = S - E$, the complementary event of *E*, is given by

$$p(\overline{E}) = 1 - p(E).$$

Proof: Using the fact that $|\bar{E}| = |S| - |E|$,

$$p(\overline{E}) = \frac{|S|-|E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E). \quad \blacktriangleleft$$

Example

What is the probability that when two dice are rolled, the sum of the numbers on the two dice is not 7?

- Let *E* be the event to roll a 7.
- This probability is 1/6.
- Thus the probability of not rolling a 7 is 1 – 1/6 = 5/6

Example

A sequence of 10 bits is chosen randomly. What is the probability that at least one of these bits is 0?

- Let *E* be the event that at least one of the 10 bits is 0.
- Then $\bar{E}$ is the event that all of the bits are 1s.
- The size of the sample space *S* is 2^{10}. Hence,

$$p(E) = 1 - p(\overline{E}) = 1 - \frac{|\overline{E}|}{|S|} = 1 - \frac{1}{2^{10}} = 1 - \frac{1}{1024} = \frac{1023}{1024}.$$

The Probability of Unions of Events

Theorem 2: Let E_1 and E_2 be events in the sample space S. Then

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

Proof: Given the inclusion-exclusion formula
$|A \cup B| = |A| + |B| - |A \cap B|$, it follows that

$$\begin{aligned} p(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} = \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|} \\ &= \frac{|E_1|}{|S|} + \frac{|E_2|}{|S|} - \frac{|E_1 \cap E_2|}{|S|} \\ &= p(E_1) + p(E_2) - p(E_1 \cap E_2). \quad \blacktriangleleft \end{aligned}$$

Example

What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?

- Let E_1 be the event that the integer is divisible by 2 and E_2 be the event that it is divisible 5.
- Then the event that the integer is divisible by 2 **or** 5 is $E_1 \cup E_2$
- The event that the integer is divisible by 2 **and** 5 is $E_1 \cap E_2$
- It follows that:

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$
$$= 50/100 + 20/100 - 10/100 = 3/5.$$

Example

When rolling two dices, what is the probability to roll a 6?

- Let E_1 be the event that the first dice is a 6 and E_2 be the event that the second dice is a 6.
- Then the event that one dice is a 6 is $E_1 \cup E_2$
- The event that both dices are a 6 is $E_1 \cap E_2$
- It follows that:

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2) = 1/6 + 1/6 - 1/36 = 11/36$$

Summary

- Complements
- Unions
- Examples

Probability Theory

Section 7.2

Video 75: Probability Distributions

- Limitations of Laplace’s Probability
- Probability Distribution
- Laws on Probability Distributions

Limitation of Laplace's Definition

Laplace's definition from the previous section, assumes that all outcomes are equally likely. This is not always the case.

Example: The sum of rolling two dices.

Set of possible outcomes: S = {2,3,4,5,6,7,8,9,10,11,12}

Probability of sum 2: need to roll (1,1), thus 1/36

Probability of sum 3: need to roll (1,2) or (2,1), thus 1/18

Probability of sum 4: need to roll (1,3), (2,2) or (3,1), thus 1/12

Probability Distribution

Definition: Let S be a sample space of an experiment with a finite or countable number of outcomes. We assign a probability $p(s)$ to each outcome s, so that:

1. $0 \leq p(s) \leq 1$ for each $s \in S$
2. $\sum_{s \in S} p(s) = 1$

The function p from the set of all outcomes of the sample space S is called a **probability distribution**.

Example

The sum of rolling two dices.

Set of possible outcomes: S = {2,3,4,5,6,7,8,9,10,11,12}

The Probability Distribution is given by:

$$p(7-i) = \frac{6-|i|}{36} \quad \text{for } -5 \leq i \leq 5$$

Example

What probabilities should we assign to the outcomes H(heads) and T (tails) when the coin is biased so that heads come up twice as often as tails?

For a biased coin, we have $p(H) = 2p(T)$.

Because $p(H) + p(T) = 1$, it follows that $2p(T) + p(T) = 3p(T) = 1$.

Hence, $p(T) = 1/3$ and $p(H) = 2/3$.

Uniform Distribution

Definition: Suppose that S is a set with n elements. The **uniform distribution** assigns the probability $1/n$ to each element of S.

Example: Consider again the coin flipping example, but with a fair coin.

Then $p(H) = p(T) = 1/2$.

Probability of an Event

Definition: The **probability** of the event *E* is the sum of the probabilities of the outcomes in *E.*

$$p(E) = \sum_{s \in E} p(s)$$

Note: no assumption is being made about the distribution.

Example

Suppose that a dice is biased so that 3 appears twice as often as each other number, but that the other five outcomes are equally likely.

What is the probability that an odd number appears when we roll this die?

- We want the probability of the event $E = \{1,3,5\}$.
- We have $p(3) = 2/7$ and $p(1) = p(2) = p(4) = p(5) = p(6) = 1/7$
- Hence, $p(E) = p(1) + p(3) + p(5) = 1/7 + 2/7 + 1/7 = 4/7$

Probabilities of Complements and Unions

Complements: $p(\bar{E}) = 1 - p(E)$ still holds.

Proof:

$$p(\bar{E}) = \sum_{s \in \bar{E}} p(s) = \sum_{s \in S - E} p(s) = \sum_{s \in S} p(s) - \sum_{s \in E} p(s) = 1 - p(E)$$

Union: $p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$ still holds.

Combinations of Events

Theorem 3: If E_1, E_2, … is a sequence of pairwise disjoint events in a sample space S, then

$$p\left(\bigcup_i E_i\right) = \sum_i p(E_i)$$

Summary

- Limitations of Laplace’s Probability
- Probability Distribution
  - Uniform distribution
  - Probability of events
- Laws on Probability Distributions
  - Complement and Union
  - Disjoint events

Video 76: Conditional Probability

- Conditional Probability
- Independence

Conditional Probability

Often probabilities exist in some context, or when a certain condition is satisfied:

- what’s chance to test positive on Corona?
- what’s chance to test positive on Corona if I feel sick?
- what’s the chance of having heads if the last 5 tosses were tails?

Generally speaking: intuition cannot be trusted

Conditional Probability

Definition: Let *E* and *F* be events with *p(F)* > 0. The **conditional probability** of *E* given *F*, denoted by *p*(*E*|*F*), is defined as:

$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

It can be interpreted as the probability that *E* occurs **given the fact (or knowing)** that *F* occurs.

Example

When you roll a dice, what's probability outcome is even?

- Without any additional knowledge: 3/6 = 1/2
- If we know that the dice is ≤3 the probability becomes 1/3

since $F = \{1,2,3\}$

$p(F) = 1/2$

$E = \{2,4,6\}$

$E \cap F = \{2\}$

$p(E \cap F) = 1/6$

$p(E|F) = p(E \cap F)/p(F) = (1/6)/(1/2) = 1/3$

Example

Tossing a coin 6 times. What is probability that the last toss is heads?

- Without any additional knowledge: 1⁄2
- But given that first five tosses are tails?

F = {tttttt,ttttth}

$p(F)$ = 2/2^6 = 1/2^5

$E = E \cap F$ = {ttttth}

$p(E \cap F)$ = 1/2^6

$p(E|F) = p(E \cap F)/p(F)$ =(1/2^6)/ (1/2^5) = 1⁄2

probability remains 1⁄2

An additional condition **may or may no**t affect probability!

Independence

Definition: The events *E* and *F* are **independent** if and only if

$$p(E \cap F) = p(E)\, p(F)$$

Theorem 4: If E and F are independent, then $p(E|F) = p(E)$

Proof: $p(E|F) = p(E \cap F)/p(F) = p(E)\, p(F)/p(F) = p(E)$

Example

Assume that each of the four ways a family can have two children {BB, GG, BG, GB} is equally likely.

Are the events *E*, that a family with two children has both girls and boys, and *F*, that a family with two children has at most one boy, independent?

- E = {BG, GB} thus $p(E) = 1/2$.
- F = {GG, BG, GB}, $p(F) = 3/4$ and $p(E \cap F) = 1/2$.
- Since $p(E)\, p(F) = 3/8 \neq 1/2 = p(E \cap F)$

  the events *E* and *F* are **not independent**

Example

Assume that each of the 8 ways a family can have three children {BBB, BBG, GGG, GGB, BGB, BGG, GBB, GBG} is equally likely.

Are the events *E*, that a family with three children has both girls and boys, and *F*, that a family with three children has at most one boy, independent?

- E = {BBG, GGB, BGB, BGG, GBB, GBG}, $p(E) = 6/8$.
- F = {GGB, BGG, GBG, GGG}, $p(F) = 4/8$ and $p(E \cap F) = 3/8$.
- Since $p(E)\, p(F) = 24/64 = 3/8 = p(E \cap F)$

  the events *E* and *F* are **independent.**

Intuition on independence of events can be deceiving!

Pairwise and Mutual Independence

Definition: The events $E_1, E_2, \ldots, E_n$ are **pairwise independent** if and only if $p(E_i \cap E_j) = p(E_i)\, p(E_j)$ for all pairs i and j with $i \leq j \leq n$.

The events are **mutually independent** if

$$p(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_m}) = p(E_{i_1})p(E_{i_2}) \cdots p(E_{i_m})$$

whenever i_j, $j = 1,2,\ldots, m$, are integers with

$1 \leq i_1 < i_2 < \cdots < i_m \leq n$ and $m \geq 2$.

- Mutual Independence implies pairwise independence

Example

Toss a fair coin twice.

E_1 = first toss is 1, $p(E_1) = ½$

E_2 = second toss is 1, $p(E_2) = ½$

E_3 = the two outcomes are different, $p(E_3) = ½$

E_1, E_2, E_3 are pairwise independent, e.g. $p(E_1 \cap E_3) = ¼ = p(E_1)\, p(E_3)$

But $p(E_1 \cap E_2 \cap E_3) = 0 \neq p(E_1)\, p(E_2)\, p(E_3) = 1/8$

- Pairwise independence does not imply mutual independence!

Summary

- Conditional Probability
- Independence
  - Independence of two events
  - Independence of multiple events

Video 77: Bernoulli Trials

- Bernoulli trials
- Binomial distribution

Example

A coin is biased so that the probability of heads is 2/3.

What is the probability that exactly four heads occur when the coin is flipped seven times?

- There are $2^7 = 128$ possible outcomes.
- The number of ways four of the seven flips can be heads is $C(7, 4)$.
- The probability of each of the outcomes is $(2/3)^4 (1/3)^3$ since the seven flips are independent.
- Hence, the probability that exactly four heads occur is

$$C(7,4)\,(2/3)^4\,(1/3)^3 = (35 \cdot 16)/2^7 = 560/2187$$

Bernoulli Trials

James Bernoulli
(1854 – 1705)

Definition: Given an experiment that can have only two possible outcomes.

- Each performance of the experiment is called a **Bernoulli trial**.
- One outcome is called a **success** and the other a **failure**.
- If p is the probability of success and q the probability of failure, then $p + q = 1$.

Frequent question: determine the probability of k successes when an experiment consists of n mutually independent Bernoulli trials.

Independent Bernoulli Trials

Theorem 5: The probability of exactly k successes in n independent Bernoulli trials, with probability of success p and probability of failure $q = 1 - p$, is

$$C(n,k)\, p^k q^{n-k}.$$

Proof: The outcome of n Bernoulli trials is an n-tuple $(t_1,t_2,\ldots,t_n)$, where each is t_i either S (success) or F (failure).

The probability of each outcome of n trials consisting of k successes and $n - k$ failures (in any order) is $p^k q^{n-k}$.

Because there are $C(n,k)$ n-tuples of Ss and Fs that contain exactly k Ss, the probability of k successes is $C(n,k)\, p^k q^{n-k}$.

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Binomial Distribution

We denote by $b(k{:}n, p)$ the probability of k successes in n independent Bernoulli trials with p the probability of success.

Viewed as a function of k, $b(k{:}n,p)$ is the **binomial distribution**.

By the previous Theorem 5,

$$b(k{:}n, p) = C(n,k)\, p^k\, q^{n-k}.$$

Summary

- Bernoulli trials
- Binomial distribution

Bayes' Theorem

Section 7.3

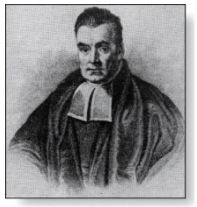
Video 78: Bayes' Theorem

- Bayes' Theorem
- Generalized Bayes' Theorem

Motivation for Bayes' Theorem

- Bayes' theorem allows us to use probability to answer questions such as the following:
  - Given that someone tests positive for having Corona, what is the probability that they actually do have Corona?
  - Given that someone tests negative for Corona, what is the probability, that in fact they do have Corona?
- Bayes' theorem has applications to medicine, law, artificial intelligence, engineering, and many diverse other areas.

Thomas Bayes
(1702-1761)



Bayes' Theorem

Bayes' Theorem: Suppose that *E* and *F* are events from a sample space S such that $p(E) \neq 0$ and $p(F) \neq 0$. Then:

$$p(F|E) = \frac{p(E|F)p(F)}{p(E|F)p(F) + p(E|\bar{F})p(\bar{F})}$$

or alternatively (a more common formulation)

$$p(F|E) = \frac{p(E|F)p(F)}{p(F)}$$

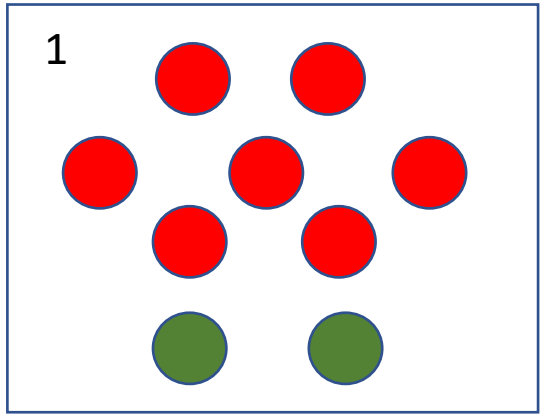
Example

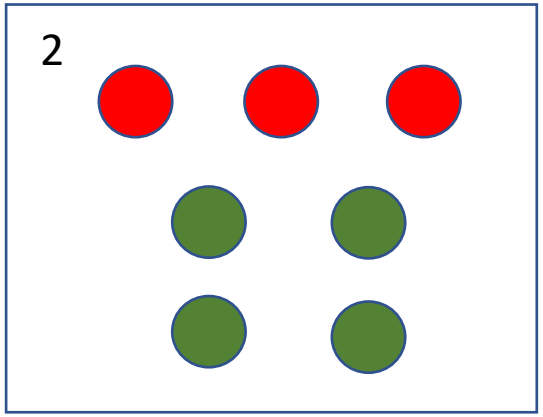
We have two boxes. The first box contains two green balls and seven red balls. The second contains four green balls and three red balls.

Bob selects one of the boxes at random.

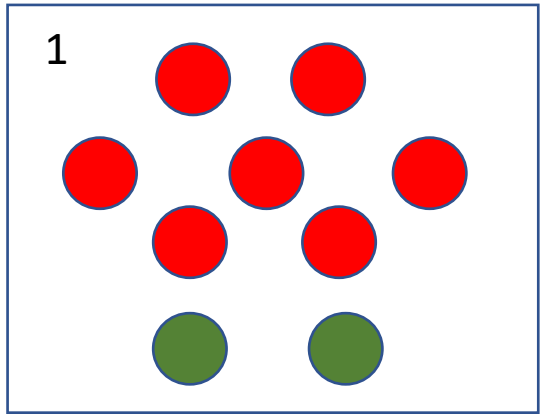
Then he selects a ball from that box at random.

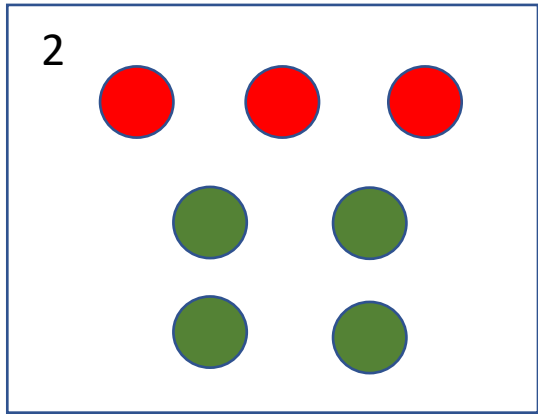
If he has a red ball, what is the probability that he selected a ball from the first box?





Example





- Let *E* be the event that Bob has chosen a red ball and *F* be the event that Bob has chosen the first box.
- $p(E|F) = \frac{7}{9}$ if he chooses the first box and has a red ball
- $p(E|\bar{F}) = \frac{3}{7}$ if he chooses the second box and has a red ball
- $p(F) = p(\bar{F}) = {}^{1}/_{2}$ as he chooses the boxes at random.

Therefore by Bayes' theorem the probability that Bob has picked the first box is:

$$p(F|E) = \frac{p(E|F)p(F)}{p(E|F)p(F) + p(E|\bar{F})p(\bar{F})} = \frac{(\frac{7}{9})(\frac{1}{2})}{\left(\frac{7}{9}\right)\left(\frac{1}{2}\right) + (\frac{3}{7})(\frac{1}{2})} = \frac{7/18}{38/63}$$

$$\approx 0.645$$

Derivation of Bayes' Theorem

Using the definition of the conditional probability $p(E|F)$ and $p(F|E)$:

$$p(E|F) = \frac{p(E \cap F)}{p(F)}, p(F|E) = \frac{p(E \cap F)}{p(E)}$$

therefore $p(E|F)p(F) = p(E \cap F)$ and $p(F|E)p(E) = p(E \cap F)$

and $p(E|F)p(F) = p(F|E)p(E)$

from which we get

$$p(E|F) = \frac{p(F|E)p(E)}{p(F)} \text{ and } p(F|E) = \frac{p(E|F)p(F)}{p(E)}$$

Derivation of Bayes' Theorem

Note that $p(E) = p(E|F)p(F) + p(E|\bar{F})p(\bar{F})$
since $(E \cap F) \cap (E \cap \bar{F}) = \emptyset$ and $(E \cap F) \cup (E \cap \bar{F}) = E$
By Theorem 3 we have

$$p(E) = p\big((E \cap F) \cup (E \cap \bar{F})\big) = p(E \cap F) + p(E \cap \bar{F})$$

Using $p(E|F)p(F) = p(E \cap F)$ and $p(F|E)p(E) = p(E \cap F)$
we get $p(E) = p(E|F)p(F) + p(E|\bar{F})p(\bar{F})$
and from $p(F|E) = \frac{p(E|F)p(F)}{p(E)}$
we have

$$p(F|E) = \frac{p(E|F)p(F)}{p(E|F)p(F) + p(E|\bar{F})p(\bar{F})}$$

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Example

Suppose that 2 persons in 100 have a Covid-19. There is a test for the disease that gives a positive result 95% of the time when given to someone with the disease. When given to someone without the disease, 98% of the time it gives a negative result. Find

a) the probability that a person who test positive has Covid-19.
b) the probability that a person who test negative does not have Covid-19.

Should someone who tests positive be worried?

Example

Let *D* be the event that the person has the disease, and *E* be the event that this person tests positive.

We know that

$$p(D) = 0.02, p(E|D) = 0.95, p(E|\bar{D}) = 0.02$$

We want to know p(D|E)

$$p(D|E) = \frac{p(E|D)p(D)}{p(E|D)p(D)+p(E|\bar{D})p(\bar{D})} = \frac{0.95*0.02}{0.95*0.02+0.02*0.98} \approx 0.49$$

So a positive test means you have about 50% chance to have the disease!

Example

And if the result is negative?

$$p(D|\bar{E}) = \frac{p(\bar{E}|D)p(D)}{p(\bar{E}|D)p(D)+p(\bar{E}|\bar{D})p(\bar{D})} = \frac{0.05*0.02}{0.05*0.02+0.98*0.98} \approx 0.001$$

So a negative test means you have about 99.9% chance to not have the disease!

Generalized Bayes' Theorem

Generalized Bayes' Theorem: Suppose that *E* is an event from a sample space *S* and that $F_1, F_2, \ldots, F_n$ are mutually exclusive events such that

$$\bigcup_{i}^{n} F_i = S.$$

Assume that $p(E) \neq 0$ for $i = 1, 2, \ldots, n$. Then

$$p(F_j|E) = \frac{p(E|F_j)p(F_j)}{\sum_{i=1}^{n} p(E|F_i)p(F_i)}.$$

Summary

- Bayes' Theorem
  - Testing diseases
- Generalized Bayes' Theorem